

M. PHARM
(SEM I) THEORY EXAMINATION 2022-23
PHARMACOLOGICAL AND TOXICOLOGICAL SCREENING METHODS-I

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief. 2 x 10 = 20**

- (a) State the role of IAEC.
- (b) Define the terms “anesthesia” and “euthanasia”
- (c) Mention the *in-vitro* models for screening anti-cancer drugs.
- (d) Enlist the preclinical screening models for the evaluation of antiallergics.
- (e) Discuss the phases of the estrous cycle of rat.
- (f) Recall some models for screening antiparkinsonian agents.
- (g) Write the applications of an immunoassay.
- (h) Enlist the *in-vivo* and *in-vitro* models for screening parasympathomimetics.
- (i) Categorize models for screening anti-diabetic drugs.
- (j) Mention *in-vivo* and *in-vitro* screening models of immunomodulators.

SECTION B**2. Attempt any two parts of the following: 10 x 2 = 20**

- (a) Discuss the methods for evaluating central and peripheral analgesics.
- (b) Elaborate on various methods of immunoassays.
- (c) Illustrate the preclinical screening models for the evaluation of anti-Alzheimer drugs.

SECTION C**3. Attempt any five parts of the following: 7 x 5 = 35**

- (a) Describe the requirements of Good lab Practices.
- (b) Define transgenesis. Write a detailed account on the methods of production of transgenic animals.
- (c) Explain the screening models for the evaluation of anti-ulcer drugs.
- (d) Illustrate the preclinical models for screening antiarrhythmic drugs.
- (e) Discuss the preclinical screening methods of anti-anxiety agents.
- (f) Describe hepatoprotective screening methods.
- (g) Summarize the limitations of animal experimentation and highlight some alternate animal experiments.